Reviewer 1

Comments

Reviewer 1 - Initial submission, 30/06/2026
Overall evaluation: The manuscript is well-written, and the research topic is an interesting and important extension of the existing auditory adaptation literature. The experimental design effectively draws on previous paradigms in face adaptation and translates it into a practical tool to study whether voice emotion adaptation is fine-tuned to speaker and speech. I only have some minor comments on the interpretation of results, clarity, and consistency of the manuscript.
Specific comments:
Page 2, Abstract, fifth-to-last line: “Indeed, the hypothesized pattern was observed for speaker gender and pseudoword.” From the main text, it seems that the hypothesized pattern was observed for speaker gender, but not pseudoword.
Page 5, sixth-to-last line: it could be added that auditory adaptation is not limited to extra-linguistic information. It happens at multiple levels of voice processing, including phoneme perception (e.g., Holt, 2006, https://doi.org/10.1121/1.2195119) and categorizing sounds into human voice and non-voice (Gao & Oxenham, 2025, https://doi.org/10.1037/xge0001672). Also, auditory adaptation is likely a domain-general effect, since it has been observed in the categorization of musical instruments as well (Lanning & Stilp, 2020, https://doi.org/10.3758/s13414-020-01980-w).
Page 7, line 7: it might be worth mentioning that the auditory simultaneous opposite adaptation paradigm presented the adaptors sequentially, not simultaneously.
Page 15: for the paired t-tests, were the p-values adjusted for multiple comparisons?
Page 18, “2.4 Interim summary”: At this point, it is still unknown whether the observed effect is due to simultaneous opposite aftereffects for speaker gender or speaker identity. Since the adaptor and target stimuli were created from recordings by the same four speakers, the adaptor-target pairs that share from the same speaker identity always have the same speaker gender. In other words, the result patterns observed in Experiment 1 could be anywhere between “fully attributable to speaker gender effect” and “fully attributable to speaker identity effect”. This could be mentioned in section 2.4. Combining results from both Experiments 1 and 2, it can be concluded that the result patterns observed in Experiment 1 are primarily driven by speaker gender, not speaker identity.
Page 35, last paragraph: this is a very valid interpretation. Another possible contributing factor is the acoustical similarity between the voice cues – different speakers of the same gender may generally have a higher degree of acoustic similarity compared to speakers of different genders, even if the listener is familiar with all speakers and their identities can be easily identified. It is possible that both classification ambiguity and acoustic similarity contribute to the conflicting pattern for speaker identity in Experiment 2.
Page 36, section 5.2, line 3: I might have missed it, but I did not find Bayesian inference analysis reported in the Results section for Experiment 3.

Reviewer 2

Comments

Reviewer 2 - Initial submission, 26/07/2026
My main concern of this study is on how the speakers were selected in the first place - the

key manipulation of this study relies on the features related to the specific emotions (i.e., fear, anger), but no information was provided on how the original speech sounded. As you have noted in p.28 ln 3-5, "M3 was perceived far more often as fearful and M4 more often as angry", which raises the question whether and how each speaker conveyed their emotions equally effective. You could have recruited another batch of listeners for evaluating speakers before the adaptation experiment, such that speakers for both genders sounded equally emotional for both emotions of interest. Also, it is recommended to use professional actors from the same school/theatre, as they often receive the same training and are capable of producing emotional speech in a very controlled manner.

Some minor suggestions below:

p.6 ln.56 "potentially based on stereotypical beliefs shared by listeners" -> Is there any evidence for this?
p.7 ln.32: You talked about "linguistic cues", but could you add some review of the roles of linguistic cues in emotional adaptation in section 1.3? Your literature review so far seems to focus solely on the extralinguistic cues.
p.8 ln.16: The identity example you gave here seems to be related to the sports team a fan supports, but in your experiment the "identity" seems to be tied to the specific person themselves. Consider change your example or simply explain here what "identity" means in your context.
p.9 ln.47: "...without hearing impariments" - how did you screen this? By interview or a hearing test?